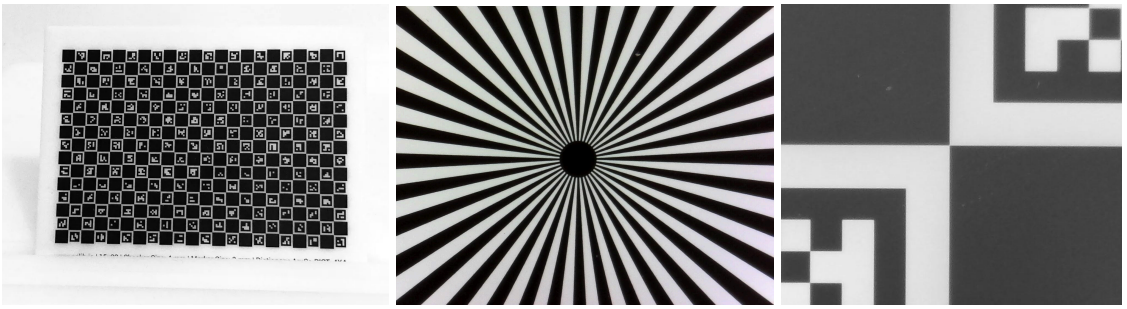
Chrome-on-Ceramic Lithographic Calibration Targets



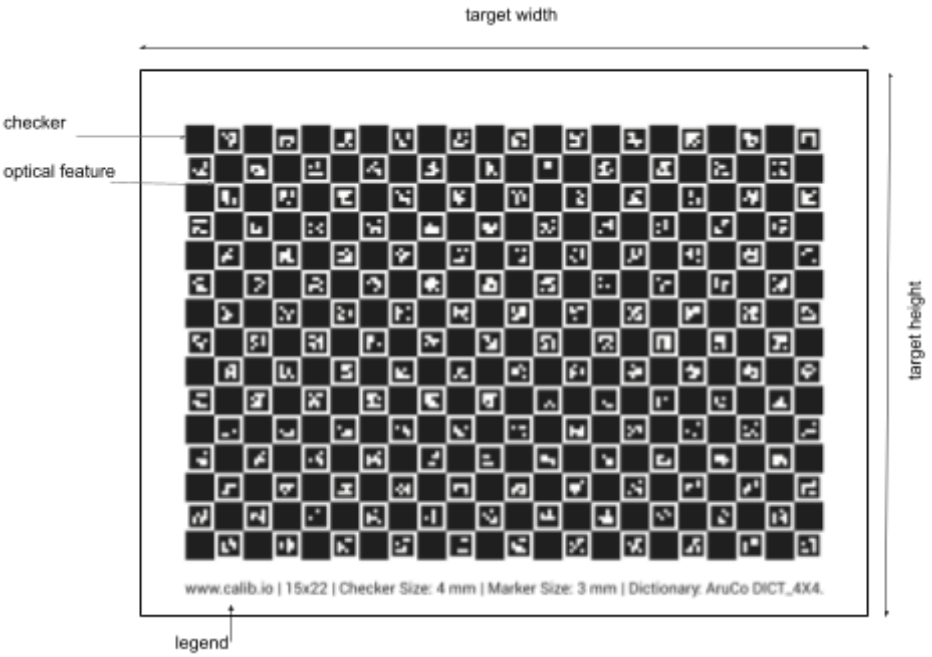
Optical Specifications

Patterning Process	Chrome etched lithography
Feature Distance Tolerance	0.005mm + 0.010mm/m
Flatness Tolerance	0.01mm per 100mm
Resolution	100000 dpi laser exposure
Gloss (@60deg)	3 (black) / 3 (white)[1]
Optical density	Very high (black) / translucent (white)
Illumination (recommended)	260 - 1400nm frontlit or backlit

Mechanical Specifications

Maximum Dimensions	400 x 300mm
Thickness	0.6 - 6mm
Weight	16.64 kg/m² at 6mm thickness
Temperature Resistance	-10 °C to +200 °C [14 °F to +392 °F]
Specified temperature	20 °C (ISO 1:2016)
Linear thermal expansion	~3 µm /(°C · m)

[1] For small feature targets, gloss values are higher due to manufacturing constraints.



Note: pattern position on target is not toleranced.

Substrate Properties

Ceramic etched lithography targets offer the ultimate in terms of dimensional accuracy and thermal stability.

The modified ceramic substrate is sputter-coated with a low gloss blue chrome layer. The pattern is transferred using direct imaging lithography, and etched. This combination of substrate and patterning process is ideal for metrology and lab use, and for high end factory calibration. Custom patterns can be created, but incur startup cost and approximately two weeks turnaround time.

For applications requiring larger targets, or complete opaqueness of the white background, we recommend lithographic printed targets or UV precision printed targets.

Dimensional tolerances

Ceramic lithography targets are manufactured on high end lithographic machines in a clean room environment. The manufacturing tolerance on this process is 0.005mm + 0.010mm/m, i.e. for two optical features spaced 1m apart, the maximum permissible distance error is 0.015mm.

Customization

Lithographic targets are manufactured to customer requirements. Lead times are generally between 2 and 3 weeks.

- Custom patterns:
  - Minimum 0.5mm margin to target edges.
  - Provided as 100% scale PDF complete vector graphics.
- Customized legend text and/or logos.
- Drawing services can be offered:
  - ChAruCo targets with different dictionaries and/or marker ids.
  - Circle/checkerboard targets with unique finder patterns
  - Deltille grid patterns and other less common targets.
  - Single AruCo marker plates for VR and robot calibration purposes.
  - Resolution test charts.

Inspection and certification

Calibration targets are always carefully inspected before shipping.

The following additional certifications can be purchased:

1. Calib.io Certification. We perform several distance and flatness measurements and certify that our manufacturing and product specifications are being met.

2. Measurement Report: The absolute 3D position of each saddle point is measured and provided in a measurement report and as a digital file. For targets up to 400mm, a calibrated optical coordinate measurement machine (OMM) is used in an ISO-17025 accredited measurement laboratory. For larger targets, we use a custom measurement setup.

Please contact us for a quote for these certification options